Cape Peninsula University of Technology

Artificial Intelligence 5 (AIN58XS) – Tutorial 2

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Every solution must be achieved programmatically. The code and output for each question must be placed in a report or a Jupyter notebook can be printed.

1. Read a local file called “wfa_girls_0-to-13-weeks_zscores.xlsx" into a dataframe. (See section 4 in “ch6&9primer” notebook)
2. Look at the top and bottom 5 rows of the data using head() and tail().
3. Plot Week vs SD0. (See section 3 in “ch6&9primer” notebook)
4. Assign suitable axis labels to the plot.
5. Plot day vs SD0, SD1, and SD1neg. (Hint: you will need to derive the days from the weeks by doing a simple scaling of the week value, e.g., week x days in a month)
6. Save the plot as an image. The format of the image is unimportant, experiment with the options.
7. Scale any SD column by a factor of 100. (See section 5.1 and 5.2 in “ch05(notes)v3” notebook)
8. Change all the column headings to lower case.
9. Sort all data by descending week.
10. Find the index and value of the min, mean, and max values in the S column. (See section 5.3 and 5.3 in “ch5” notebook)
11. Calculate the percent change of any SD column and save the result as a new dataframe column named "PCSDx".
12. Write the dataframe to a file named “tutorial2.csv”.
13. Confirm the image and csv files you generated are as you expect.
14. Export your notebook to pdf. (This step is important for tests)